

ESB 2LBC-BI 5

MODELISATION — Academic Year 2025-2026



Pizza Sales

Business Intelligence Project

Institution	ESB
Academic Year	2025 - 2026
Subject	Business Intelligence
Profesor	Mrs Dalila Amara
Group Members	Raed Meddeb
	Wajdi Riahi
	Noureddine Chehimi

Tools Used	PostgreSQL · PgAdmin 4 · Talend Open Studio · Power BI
Dataset	Pizza Place Sales — 48,620 order records (2015)
Architecture	Source DB → ETL → Data Warehouse → Dashboard

1. Introduction & Objective

1.1 Context and Motivation

This project was carried out as part of the Business Intelligence with the aim of applying the key concepts studied throughout the semester: star schema modeling, ETL processes, data warehouse design, and interactive dashboard creation.

The dataset chosen for this project comes from a real pizza restaurant operating in 2015. It contains detailed transactional data covering an entire year of sales activity, making it ideal for multi-dimensional analysis.

1.2 Research Problem

How can a pizza restaurant optimize its sales strategy by analyzing ordering patterns across time, pizza types, and order behavior to maximize revenue and improve business decisions?

1.3 Objectives

- Build a complete data warehouse based on a star schema model
- Implement an automated ETL pipeline using Talend Open Studio
- Load and transform data from a PostgreSQL source database into a data warehouse
- Create interactive dashboards in Power BI for business analysis
- Answer key business questions through data visualization

1.4 Dataset Overview

The dataset consists of 4 CSV source files from a pizza restaurant for the year 2015:

File	Records	Key Columns	Role
orders.csv	21,350	order_id, date, time	When orders were placed
order_details.csv	48,620	order_details_id, order_id, pizza_id, quantity	What was ordered
pizzas.csv	96	pizza_id, pizza_type_id, size, price	Pizza variants and prices
pizza_types.csv	32	pizza_type_id, name, category, ingredients	Pizza types and categories

2. Tools Used & Data Modeling

2.1 Tools

Tool	Version/Type	Role in Project
PostgreSQL + PgAdmin 4	PostgreSQL 16	Source database and data warehouse hosting
Talend Open Studio	Data Integration	ETL — Extract, Transform, Load
Power BI Desktop	Microsoft Power BI	Data visualization and dashboard creation

2.2 Star Schema Design

The data warehouse was designed using a star schema with one central fact table surrounded by three dimension tables. Special attention was given to the Date dimension as required by the project rules.

Table	Type	Columns
fact_sales	Fact Table	fact_id (PK), order_details_id, order_id (FK), date_id (FK), pizza_id (FK), quantity, total_price
dim_date	Dimension — TIME	date_id (PK), full_date, day, month, month_name, quarter, year, day_of_week
dim_order	Dimension	order_id (PK), order_date
dim_pizza	Dimension	pizza_id (PK), pizza_type_id, name, category, size, price, ingredients

Relationships (Foreign Keys)

- fact_sales.date_id → dim_date.date_id
- fact_sales.order_id → dim_order.order_id
- fact_sales.pizza_id → dim_pizza.pizza_id

2.3 PgAdmin — Database Setup

Two PostgreSQL databases were created using PgAdmin 4:

pizza_source: Operational database — stores the 4 raw CSV tables as imported source data

pizza_dw: Data warehouse — stores the star schema tables populated by Talend ETL jobs

3. ETL Process — Talend Open Studio

The ETL (Extract, Transform, Load) process was fully automated using Talend Open Studio for Data Integration. Four separate jobs were created, one for each dimension table and one for the fact table.

3.1 Talend Jobs Overview



Figure 1 — The 4 Talend ETL jobs created for the project

Job Name	Description	Source	Target
Load_DIM_DATE	Extracts distinct dates, derives day/month/quarter/year/day_of_week	pizza_source	dim_date
Load_DIM_ORDER	Loads all orders with their dates	pizza_source	dim_order
Load_DIM_PIZZA	Joins pizzas and pizza_types tables	pizza_source	dim_pizza
Load_FACT_SALES	Main fact job with 3 lookups, calculates total price	pizza_source + pizza_dw	fact_sales

3.2 Job Execution Results

Load_DIM_DATE — 358 rows loaded

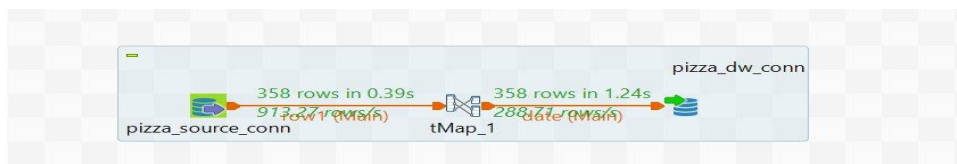


Figure 2 — Load_DIM_DATE: 358 unique dates extracted and loaded into dim_date

Load_DIM_ORDER — 21,350 rows loaded

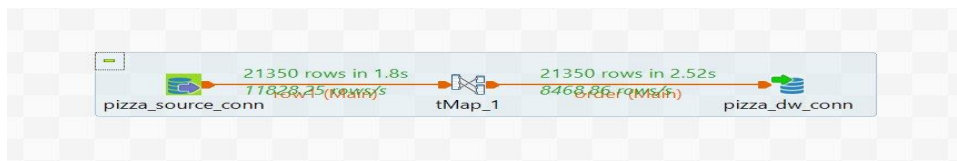


Figure 3 — Load_DIM_ORDER: 21,350 orders loaded into dim_order

Load_DIM_PIZZA — 96 rows loaded

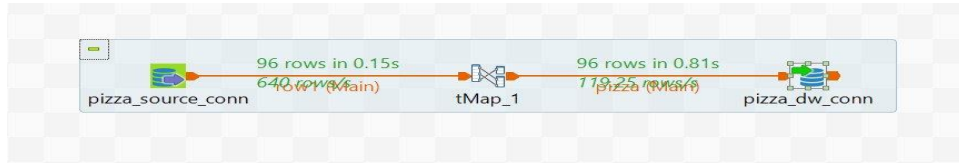


Figure 4 — Load_DIM_PIZZA: 96 pizza variants loaded into dim_pizza

Load_FACT_SALES — 48,620 rows loaded

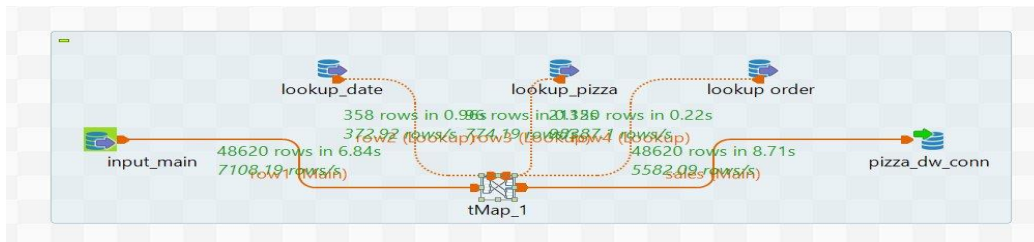


Figure 5 — Load_FACT_SALES: 48,620 fact rows with 3 lookup joins (date, pizza, order)

3.3 tMap Transformations

The tMap component was used in each job to perform transformations. Key transformations include:

- DIM_DATE: Deriving day, month, month_name, quarter, year, day_of_week from a single date field using TalendDate.formatDate()
- DIM_PIZZA: Joining pizzas and pizza_types source tables to create a unified pizza dimension
- FACT_SALES: Three inner joins — matching date via full_date, pizza via pizza_id, and order via order_id — plus calculating total_price = quantity × price

4. Power BI Dashboard & Analysis

After successfully loading the data warehouse, Power BI Desktop was connected to the pizza_dw PostgreSQL database. The star schema relationships were configured and interactive dashboards were built to answer the research questions.

4.1 Data Model in Power BI

The following relationships were set up in Power BI Model view:

- fact_sales → dim_date (Many-to-One via date_id)
- fact_sales → dim_order (Many-to-One via order_id)
- fact_sales → dim_pizza (Many-to-One via pizza_id)

4.2 DAX Measures Created

Measure	DAX Formula
Total Revenue	SUM(fact_sales[total_price])
Total Orders	DISTINCTCOUNT(fact_sales[order_id])
Total Quantity	SUM(fact_sales[quantity])
Avg Order Value	DIVIDE([Total Revenue], [Total Orders])

4.3 Dashboard

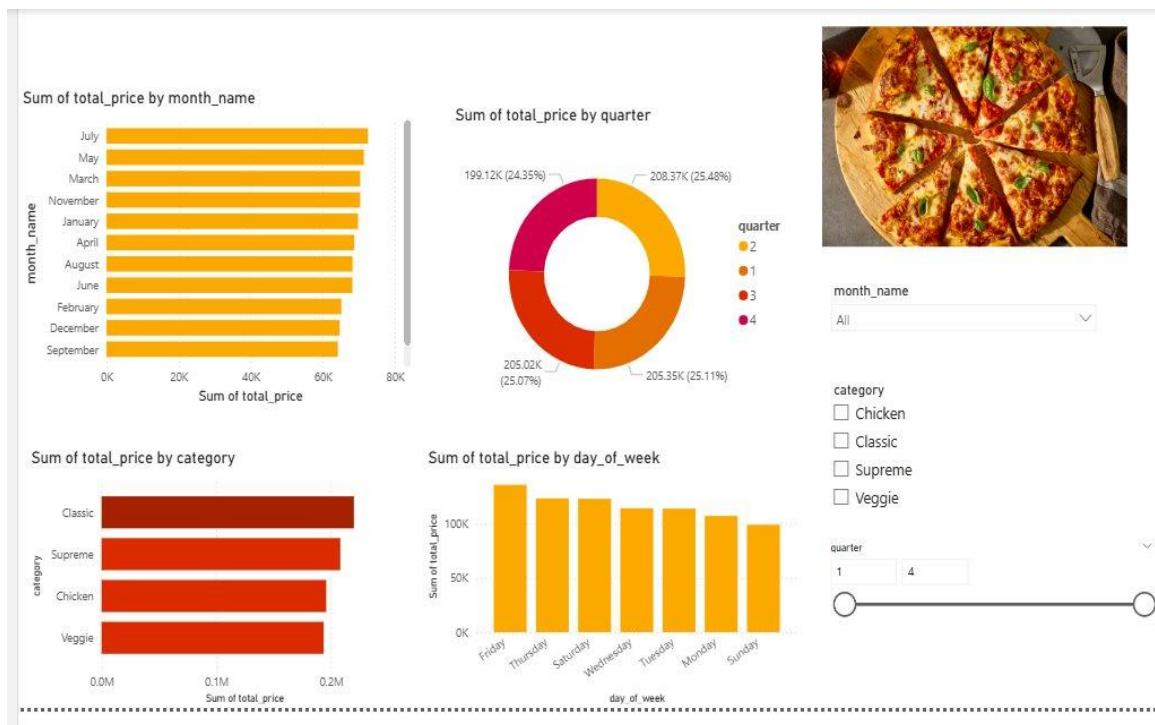


Figure 6 — Pizza Sales Power BI Dashboard with interactive slicers and multiple chart types

5. Analysis Results & Conclusion

5.1 Key Findings from the Dashboard

Revenue by Month

The monthly revenue analysis reveals that July and May are the peak revenue months, while September and October show lower sales. This suggests seasonal patterns in customer demand, with summer months driving higher pizza consumption.

Revenue by Quarter

The quarterly donut chart shows a fairly even distribution across the four quarters, with Q1 and Q2 slightly ahead. Each quarter contributes approximately 25% of total annual revenue, indicating consistent year-round demand.

Revenue by Category

The Classic pizza category generates the highest revenue, followed by Supreme, Chicken, and Veggie. This insight helps the restaurant understand which category to promote or expand.

Revenue by Day of Week

Friday is clearly the busiest day of the week, generating the highest revenue. Thursday and Wednesday follow. Sunday is the slowest day. This information is valuable for staffing and inventory planning.

5.2 Business Recommendations

- Increase marketing efforts in September and October to boost revenue during slower months
- Focus promotional offers on Friday and Thursday when customer volume is highest
- Expand the Classic pizza category as it is the top revenue driver
- Consider special offers on Sundays to improve the slowest day performance

5.3 Conclusion

This project successfully implemented a complete Business Intelligence pipeline for a pizza restaurant dataset. Starting from raw CSV files, we designed a star schema data warehouse, automated the ETL process using Talend, and built an interactive Power BI dashboard.

The project demonstrates the full BI lifecycle: from data modeling and warehouse design, through ETL transformation, to visual analytics. The resulting dashboard provides actionable insights into sales patterns, enabling data-driven business decisions.

All project requirements were met: a clearly defined research problem was addressed, the Date dimension was given special attention as required, Talend was used for automated loading, and Power BI was used for dashboarding and analysis.